Internet Banking User Satisfaction Analysis

Master's-Level Research Portfolio Case Study

Primary Survey | SPSS | R | PCA / EFA | Regression

Factors affecting users’ satisfaction with internet banking services

Sai Lao Kham • Master's-level research project

1 | Research Problem & Objectives

Business / Research Question

- Investigate the factors influencing users' satisfaction with internet banking services.
- Analyze the effects of perceived internet banking services on users' satisfaction.
- Focus on internet banking users among university staff in Monywa University of Economics and Monywa University.

Study Design

- Primary survey data
- Stratified random sampling by university
- Face-to-face interviews
- Seven-point Likert scale
- Cross-sectional study design
- Survey period: July 2020

Source: Original thesis, Chapter 1, pp. 4–5 (PDF pages 17–18).

2 | Data & Sample

316
Respondents

2
Universities

7-point
Likert scale

July 2020
Survey period

Sampling & Collection

- Target population: internet banking users among university staff.
- Stratified random sampling with university as the stratifying variable.
- Face-to-face data collection.

Study Scope

- Monywa University of Economics (MUE)
- Monywa University (MU)
- Due to time constraints, the study was limited to these two universities in Monywa.

Note: The thesis contains a sampling-count inconsistency in some sections; this portfolio uses the 316 observations reported as used for analysis.

3 | Analytical Workflow

From raw survey responses to interpretable business insights

01 Survey Design → 02 Primary Data → 03 Descriptive Statistics → 04 Reliability Testing → 05 PCA / EFA + Varimax → 06 Correlation Analysis → 07 Multiple Regression → 08 Diagnostics & Insights

Core analytical approach

• The analysis used Principal Component Analysis as the extraction method with Varimax rotation, followed by multiple linear regression using the extracted factor scores.

Source: Original thesis, Chapters 3–4.

4 | Reliability & Factorability

0.90
KMO

< 0.001
Bartlett's p-value

0.91
Service α

0.87
Satisfaction α

Factorability

- KMO = 0.90 indicated strong sampling adequacy for factor analysis.
- Bartlett's test: $\chi^2 = 3966.289$, df = 406, $p < 0.001$.
- The correlation structure was considered appropriate for factor analysis.

Internal Consistency

- Internet Banking Services: $\alpha = 0.91$ (29 items).
- Users' Satisfaction: $\alpha = 0.87$ (7 items).
- Both exceeded the 0.70 reliability benchmark used in the study.

Source: Original thesis, Chapter 4, pp. 44–45 (PDF pages 57–58).

5 | PCA: Reducing 40 Variables to 6 Factors

40
Original variables

29
Retained variables

6
Extracted factors

49.5%
Explained variance

PCA setup

• Extraction: Principal Component method • Rotation: Varimax with Kaiser normalization • Kaiser criterion: eigenvalue ≥ 1

Source: Original thesis, Chapter 4, pp. 44–48 (PDF pages 57–61).

6 | Six Underlying Factors

01
Responsiveness
7 items • α 0.84

02
Perceived Usefulness
4 items • α 0.78

03
Trust & Security
6 items • α 0.81

04
Better Services
6 items • α 0.79

05
Perceived Ease of Use
3 items • α 0.73

06
Effectiveness
3 items • α 0.76

All six factor reliabilities were above 0.70 in the thesis; cumulative variance explained = 49.5%.

7 | What the Six Factors Represent

Six factors extracted using PCA with Varimax rotation

1. Responsiveness

- Willingness to help customers
- Support, confidence and courtesy
- Knowledgeable staff
- First-time service performance

2. Perceived Usefulness

- Improves productivity
- Saves time and effort
- Useful for daily banking activities

3. Trust & Security

- Security of personal information
- Safety of online transactions
- Trust in internet banking services

4. Better Services

- Additional and after-sale services
- Faster completion of tasks
- App upgrades and maintenance
- Access to available banking services

5. Perceived Ease of Use

- Simple transaction procedures
- Easy to use internet banking
- Easy to learn and operate

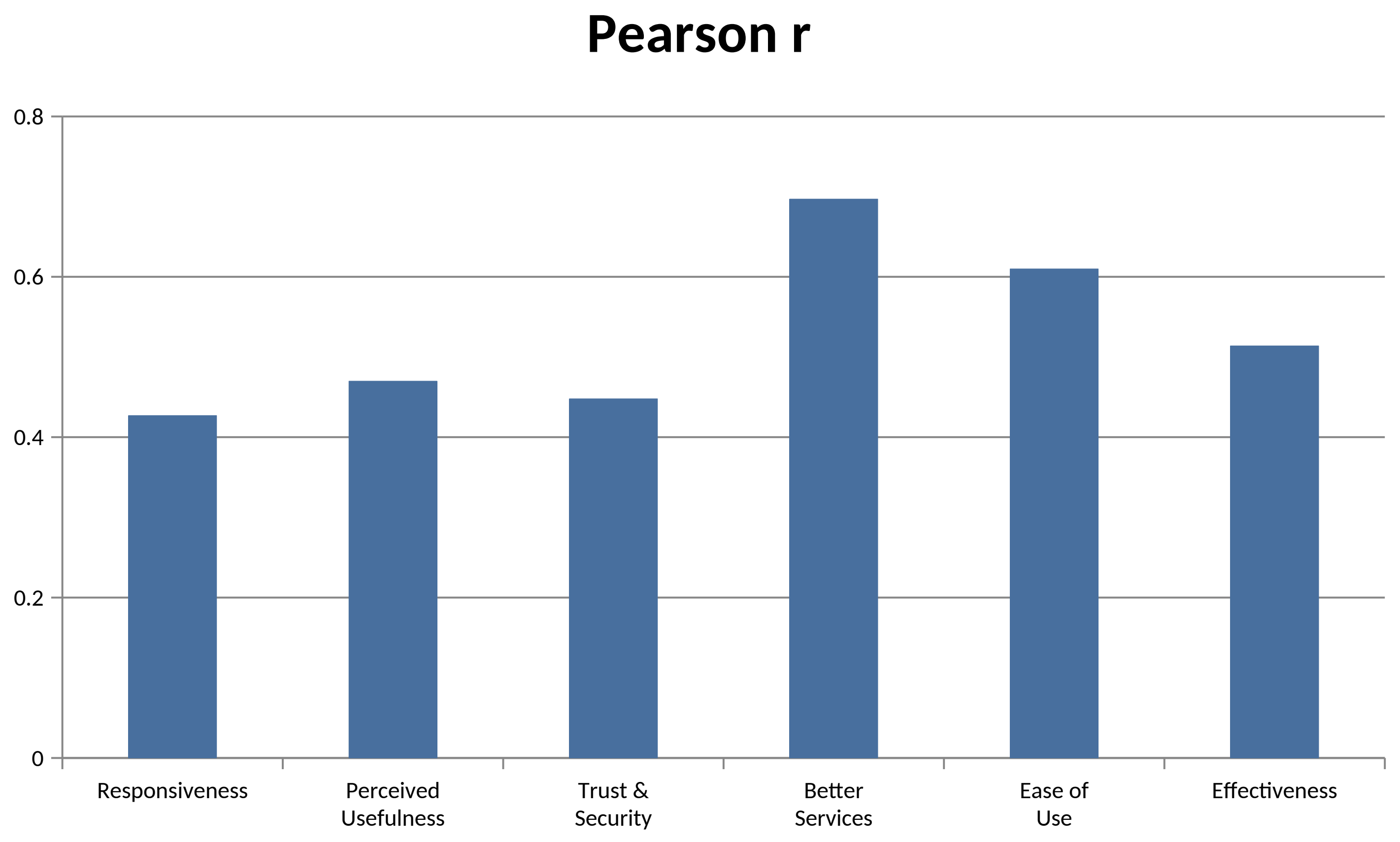
6. Effectiveness

- Good online payment system
- Effectiveness and convenience
- Better than manual / branch banking

The six-factor solution explained 49.5% of cumulative variance.

8 | Correlation with User Satisfaction

Pearson correlations reported in the original SPSS output

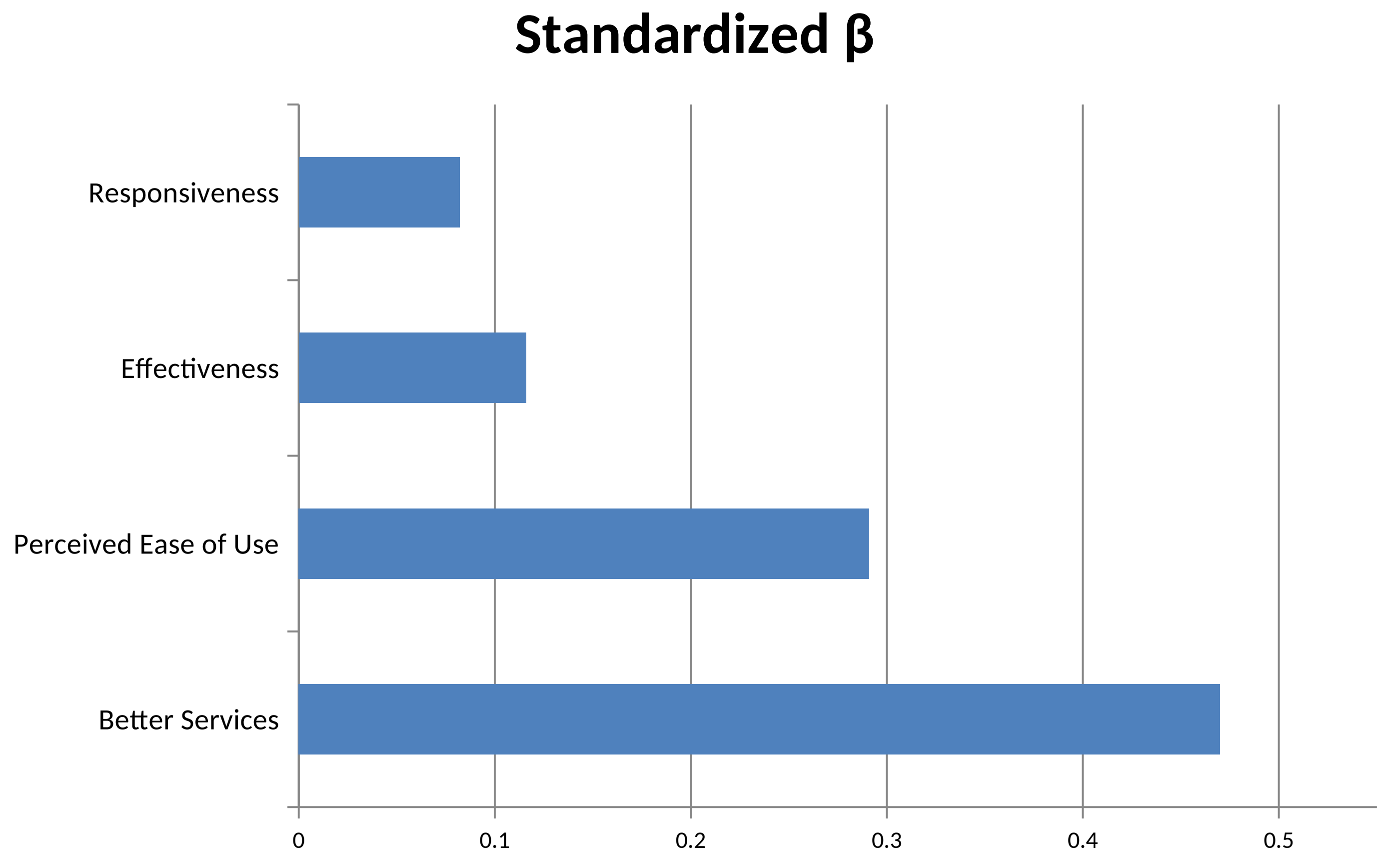


Source: Original thesis, Table 4.13 / Appendix D.

Key finding

- All six factors were positively correlated with user satisfaction (p = 0.000 reported for each).
- Strongest: Better Services (r = 0.697).
- Second strongest: Perceived Ease of Use (r = 0.610).

9 | Regression: Which Factors Matter Most?



59.4%
Adjusted R^2

116.158
F-statistic

Model significance
- Overall model: $p < 0.001$

The final regression retained four significant predictors. VIF values ranged from 1.298 to 1.586 for these predictors.

10 | Business Insights

From statistical results to management priorities

The model explains 59.4% of variation in satisfaction. The strongest actionable drivers are Better Services and Perceived Ease of Use.

1 | Service improvement is the primary lever

Better Services had the largest standardized effect ($\beta = 0.470$). Additional services, after-sale support, app maintenance and stronger online-service delivery should receive first priority.

2 | Reduce friction in the digital journey

Perceived Ease of Use was the second strongest predictor ($\beta = 0.291$). Simpler transaction procedures and a clearer online payment experience are directly relevant to satisfaction.

3 | Reliability of execution still matters

Effectiveness ($\beta = 0.116$) and Responsiveness ($\beta = 0.082$) remained significant. Customers value efficient transactions, knowledgeable support and timely assistance.

4 | Correlation is not the same as unique impact

Perceived Usefulness and Trust & Security were positively correlated with satisfaction, but were not significant in the final regression model. They should not be presented as independent drivers in this sample.

Management Priority

1. Better Services → 2. Ease of Use → 3. Effectiveness → 4. Responsiveness

Use these priorities as directional evidence from this cross-sectional study, not as causal proof.

11 | Recommendations & Next Steps

Recommendations supported by the study

• Prioritize additional and after-sale services.

• Improve online payment and transaction experience.

• Simplify digital banking procedures and user flows.

• Maintain and upgrade banking applications.

• Strengthen responsiveness and customer support.

• Continue investigating users’ concerns around safety, trust and security.

Future analytical opportunities

• Expand the sample beyond two universities and one city.

• Test the model with a larger population and sample size.

• Compare satisfaction across age, gender, occupation and other respondent groups.

• Apply the framework to other online services.

• Validate the scale in future studies.

18 | Portfolio Summary

What this project demonstrates

Analytical capabilities

- Primary survey design & data collection
- Statistical reasoning and exploratory analysis
- Reliability testing and factor reduction
- PCA / EFA with Varimax rotation
- Multiple linear regression and diagnostics
- Translation of statistical results into business insights

Tools & methods

- SPSS
- R / statistical programming
- Survey data analysis
- PCA / factor analysis
- Regression modeling
- Business / customer satisfaction analysis

Portfolio note: This document summarizes a Master's-level thesis research project and is not a claim that a degree was awarded.

13 | Original R Output: Factor Adequacy

Appendix C - KMO and Bartlett's test

R output for factor analysis

```
> library(mvnormtest)

###############################################
#Shapiro-Wilk normality test#
###############################################


data:  Z
W = 0.71447, p-value < 2.2e-16

> library(Hmisc)
> library(psych)

###############################################
#Kaiser-Meyer-Olkin factor adequacy#
###############################################

Call: KMO(r = s)
Overall MSA =  0.9
MSA for each item =
  S1   S2   S3   S4   S5   S6   S7   S9  S10  S11  S13  S14
0.86 0.93 0.91 0.86 0.90 0.90 0.88 0.91 0.91 0.88 0.92 0.92
 S15  S16  S17  S18  S20  S27  S28  S29  S30  S31  S32  S34
0.89 0.90 0.88 0.91 0.87 0.89 0.86 0.94 0.88 0.85 0.89 0.92
 S35  S36  S37  S38  S40
0.92 0.89 0.93 0.87 0.89

###############################################
#Bartlett's test of Sphericity#
###############################################

$chisq
[1] 3966.289

$p.value
[1] 0
```

Analytical Takeaway

The data were suitable for factor analysis. KMO = 0.90 indicated strong sampling adequacy, while Bartlett's test was significant ($\chi^2 = 3966.289$, df = 406, $p < 0.001$).

Decision

Proceed with PCA / factor analysis.

14 | Original R Output: Reliability & Scree Plot

Appendix C - reliability and factor-retention evidence

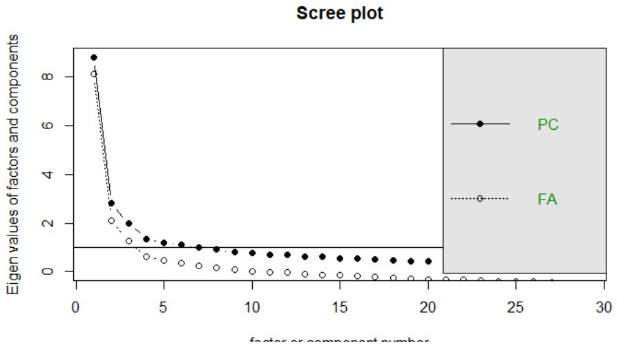
```
################################################
#Reliability analysis #
################################################

Call: alpha(x = s)

  raw_alpha std.alpha G6(smc) average_r S/N   ase mean   sd
      0.91      0.92    0.94      0.28  11 0.007  5.3 0.58
 median_r
     0.26

lower alpha upper     95% confidence boundaries
0.9 0.91 0.93

Scree of eigen values
Call: NULL
Eigen values of factors  [1]  8.12  2.10  1.25  0.62  0.48  0.37  0.2
5  0.17  0.09
[10]  0.03 -0.03 -0.05 -0.09 -0.12 -0.15 -0.18 -0.20 -0.26
[19] -0.28 -0.31 -0.33 -0.35 -0.38 -0.39 -0.40 -0.43 -0.45
[28] -0.47 -0.48
Eigen values of Principal Components [1] 8.83 2.81 1.99 1.36 1.18 1.0
9 0.98 0.92 0.80 0.75 0.70
[12] 0.69 0.62 0.60 0.54 0.53 0.51 0.46 0.45 0.42 0.40 0.36
[23] 0.34 0.32 0.31 0.28 0.26 0.24 0.23
```



Analytical Takeaway

The service-item scale showed strong internal consistency (Cronbach's α = 0.91). The eigenvalue and scree evidence supported retaining six principal components.

Result

Reliable measurement + six-factor retention.

15 | Original R Output: Rotated Factor Matrix

Appendix C - six-factor rotated solution

Rotated Factor Matrix

Variables	Factors					
	1	2	3	4	5	6
S1	0.773					
S2	0.684					
S3	0.535					
S4	0.618					
S5	0.539					
S6	0.59					
S35	0.531					
S20		0.529				
S27		0.808				
S28		0.542				
S30		0.704				
S9			0.526			
S10			0.578			
S11			0.618			
S13			0.454			
S14			0.597			
S15			0.597			
S7				0.52		
S34				0.554		
S36				0.565		
S37				0.5		
S38				0.656		
S40				0.492		
S16					0.629	
S17					0.695	
S18					0.401	
S29						0.347
S31						0.595
S32						0.658

```
               PA1   PA2   PA3   PA4   PA5   PA6
SS loadings    3.266 3.071 2.617 2.563 1.539 1.309
Proportion Var 0.113 0.106 0.090 0.088 0.053 0.045
Cumulative Var 0.113 0.219 0.309 0.397 0.450 0.495
```

Analytical Takeaway

PCA reduced the retained service variables into six interpretable factors. Together, the six factors explained 49.5% of total variance.

Result

29 retained variables → 6 underlying factors → 49.5% variance explained.

16 | Original SPSS Output: Model Fit & ANOVA

Appendix D - regression model summary and ANOVA

	MeanF3	316	316	316	316	316	316	316
	MeanF4	316	316	316	316	316	316	316
	MeanF5	316	316	316	316	316	316	316
	MeanF6	316	316	316	316	316	316	316

Model Summary[e]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.697[a]	.485	.484	.51495
2	.763[b]	.582	.580	.46455
3	.771[c]	.594	.590	.45887
4	.774[d]	.599	.594	.45668

a. Predictors: (Constant), MeanF4

b. Predictors: (Constant), MeanF4, MeanF5

c. Predictors: (Constant), MeanF4, MeanF5, MeanF6

d. Predictors: (Constant), MeanF4, MeanF5, MeanF6, MeanF1

e. Dependent Variable: MeanSatisfaction

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	78.502	1	78.502	296.041	.000[b]
	Residual	83.264	314	.265		
	Total	161.766	315			
2	Regression	94.220	2	47.110	218.300	.000[c]
	Residual	67.546	313	.216		
	Total	161.766	315			

Analytical Takeaway

The final regression model was statistically significant and explained 59.4% of the variation in user satisfaction (Adjusted $R^2 = 0.594$; $F = 116.158$; $p < 0.001$).

Model Performance

$R = 0.774$ • $R^2 = 0.599$ • $N = 316$

17 | Original SPSS Output: Significant Predictors

Appendix D - coefficients and collinearity statistics

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients			Collinearity Statistics	
		B	Std. Error	Beta	t	Sig.	Tolerance	VIF
1	(Constant)	2.251	.199		11.317	.000		
	MeanF4	.653	.038	.697	17.206	.000	1.000	1.000
2	(Constant)	1.224	.216		5.664	.000		
	MeanF4	.492	.039	.524	12.570	.000	.766	1.305
	MeanF5	.340	.040	.356	8.534	.000	.766	1.305
3	(Constant)	1.035	.223		4.647	.000		
	MeanF4	.461	.040	.492	11.525	.000	.715	1.399
	MeanF5	.288	.043	.302	6.696	.000	.640	1.562
	MeanF6	.114	.039	.131	2.966	.003	.667	1.500
4	(Constant)	.899	.232		3.875	.000		
	MeanF4	.441	.041	.470	10.710	.000	.670	1.492
	MeanF5	.278	.043	.291	6.425	.000	.630	1.586
	MeanF6	.101	.039	.116	2.611	.009	.649	1.542
	MeanF1	.071	.035	.082	1.997	.047	.771	1.298

Key Drivers of Satisfaction

Better Services (β = 0.470) was the strongest significant predictor, followed by Ease of Use (β = 0.291), Effectiveness (β = 0.116), and Responsiveness (β = 0.082).

Business Implication

Prioritize service improvement and ease of use, while strengthening effective transactions and responsive support.